14BCS0002

B.Tech Computer Engineering
5th Semester Examinations, 2016
Computer Architecture
Paper No. BCEN 501

Maximum Time: 3 hrs

Maximum Marks: 60

Note: Attempt any two parts from each question.

1. (a) What are the different performance attributes important for consideration in computer architecture and design?

(b) What are the elements of instruction set architecture?

(c) State Amdahl's Law. Using Amdahl's Law, derive the $Speedup_{overall}$ of a processor after the application of enhancement.

2. (a) Translate the following C code in the MIPS assembly language code:

```
if(a==b)
        x = x + y;
else
        x = x - y;
```

(b) Discuss the various types of delays involved in single cycle machine execution.

(c) Draw the diagram of a multi cycle microarchitecture. What is the key principle behind the multi cycle execution?

3. (a) What is flow dependence? What are the various approaches to handle flow dependences?

(b) What is scoreboarding? Explain with respect to the instruction pipelining.

(c) Define

(i) Dynamic pipeline
(ii) Collision vector
(iii) Internal data forwarding

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Total Pages:- 2

Your Roll No. 14CS0007

BCEN-505

B.tech.(Computer Engineering) V Semester Examination 2016

Microprocessor

Paper No. : CEN-505

Time: 3 Hours

Maximum Marks:60

(Write your Roll No. on top immediately on receipt of this question Paper)

Note:- Attempt all questions.

Q1. (a). Draw and explain timing diagram of STA 2250 H instruction of 8085 stored at address 2165H. 06

(b). Draw and explain Block diagram of 8085 Microprocessor. 06

2. (a) (i). Explain the mathematical function of the following program. 03

```
MVI  A, 07H
RLC
MOV B ,A
RLC
RLC
ADD B
```

(ii) Explain the different instructions available in 8085 for clearing Accumulator Register. 03

2(b) Explain the meanings of the following instructions after writing their syntax. Also illustrate their use. 06

(i)STAX

(ii)XTHL

(iii)LDA

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14BCS0002

B.Tech. (Comp. Engg.) Vth Sem Examination 2016
AUTOMATA THEORY
Paper no CEN-502

Time : Three hours **Max. Marks: 60**

Attempt all questions. Write your Roll No on top of the question paper immediately.

Q1 a) Define the Term Finite Automata and briefly explain its types with suitable example. 6 Marks

Q1 b) Construct a DFA for the followings languages and validate your answers: 6 Marks
i) ab(ba + ab + abc)*bd
ii) 1(11)*(10)*0

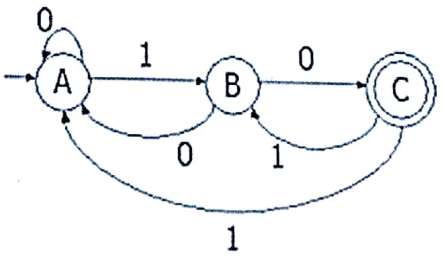
OR

Q1' a) Differentiate between the various Finite State Machines with Output using suitable example. 6 Marks

Q1' b) Construct a Mealy and a Moore Machine: 6 Marks
i) that outputs '1' whenever it receives a pair 1 in a string with Σ = (0,1).
ii) that outputs '0' whenever it receives 101 in a string with Σ = (0,1).

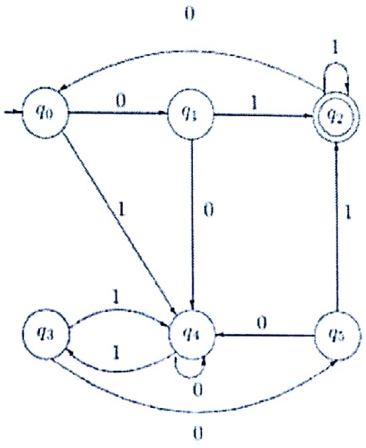
Q2 a) Using suitable reasons, verify the validity of the following statements about regular expressions. 6 Marks
i) (RS+R)*R = R(SR+R)*
ii) (RS+R)*RS = (RR*S)*

Q2 b) Convert the following NFA to DFA. 6 Marks



OR

Q2' a) Minimize the following DFA: 6 Marks



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Q2'b) Construct a regular expression for the minimum DFA obtained as an answer of Q2'(a). 6 Marks

Q3 a) Using pumping Lemma, show whether the following languages are regular or not: 6 Marks

i) $L1 = \{1^x 2^y : x, y>=0\}$

ii) $L2 = \{1^{2x} : x \text{ is odd}\}$

Give suitable reasons for each of your answer.

Q3 b) Explain the Chomsky classification of languages using suitable examples. Also construct a CFG for a language accepting all strings over {0, 1} which must contain a substring 110. 6 Marks

Q4 a) Convert the following CFG to CNF 6 Marks

$S \rightarrow AaBb \mid AaB$ $\quad A \rightarrow AB \mid a \mid \varepsilon$

$B \rightarrow CB \mid b \mid \varepsilon$ $\quad C \rightarrow ACD \mid CD \mid c \mid \varepsilon$

$D \rightarrow d$

Mention all the steps and the corresponding rules used at appropriate positions.

Q4 b) Construct a PDA for a Language that accepts set of all strings over {a, b, c} which contain equal no 'a' & 'c'. Validate your answers. Also mention the steps to convert CFG to PDA using suitable examples. 6 Marks

Q5 a) Construct a Turing Machine to identify a Language $L = \{a^n b^n c^n d^n : n >=0\}$. Also design a TM for the operation: Ц**w**Ц → Ц$\mathbf{w}^R$**w**Ц , where w^R stands for reverse of the string w. 6 Marks

Q5 b) Define Primitive Recursive functions. Show that the function $f(x, y) = 3x^2 + 2y$ is Primitive recursive. 6 Marks

OR

Q5'b) Explain the followings using suitable examples: 6 Marks

i) Recursively Enumerable Languages

ii) NP Hard Problems

iii) ε –NFA to Transition diagram conversion

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CEN-503

B. Tech. (Computer Engineering) Vth Semester Examination 2016

COMPUTER NETWORK-I

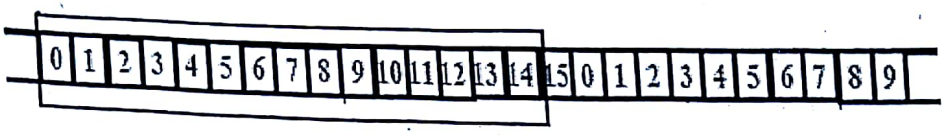
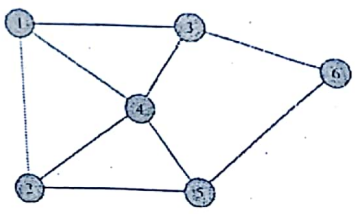
CEN-503

Time: 3 Hours **M.M: 60**

Attempt any two parts from Q. No. 1 to 4. Q. No. 5 is compulsory.

S. No.	Questions	Marks
1.	a) An ISP is granted a block of addresses with 208.200.0.0/16 (65536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: i. The first group has 64 customers: each needs 256 addresses. ii. The second group has 128 customers; each needs 128 addresses iii. The third group has 128 customers; each needs 64 addresses Design the sub blocks and find out how many addresses are still available after these allocations.	6
	b) Explain CDMA with a suitable example.	6
	c) What are the reasons for using layered protocols? What are the possible disadvantages of using layered protocols?	6
2.	a) What are the five advantages of Packet Switching over Circuit Switching?	6
	b) A CRC is constructed to generate a 4-bit FCS (Frame Check Sequence) for an 11-bit message $D = 10010011011$. The generator polynomial is $P(x)=x^4+x+1$. i. Encode the message D using the generator polynomial and give the codeword (T: Frame to be transmitted). ii. Suppose the channel introduces an error pattern $E(x)=x^{14}+x^{11}+x^{10}$ in T during the transit. What is received? Can the error be detected?	6
	c) What is p-Persistent CSMA protocol? How p-Persistent protocol behaves (in terms of collision and delay) for small value of p under heavy load and under low load?	6
3.	a) What are the assumptions for dynamic Channel Allocation?	6
	b) Explain how Collision is avoided in Wireless LAN Protocols.	6
	c) Neatly draw and explain all the fields of Typical Bluetooth data frame both basic and enhanced data frames.	6
4.	a) Why the size of window is one less than module range in sliding window protocol? Explain.	2

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	The sender windows for system uses Go-Back-n ARO are given as:  i. Show the window after the sender has sent packets 0 to 12 and has received ACK7. ii. Show the window after the sender has sent packets 0 to 14 hand has received NAK6. iii. Show the window after the sender has sent packets 0 to 14, no acknowledgement has received, and timeout has expired. iv. The receiver has sent ACK6 and ACK9 but ACK6 is lost and no timeout up to ACK9 at sender. Show the sender window.	4
	b) What are the different types of HDLC frames? Explain with their application areas.	6
	c) Sixteen stations are contending to access a shared channel in a LAN using Bit Map protocol. If the length of contention slot is 2 bits long and length of data frame is 512. Calculate the efficiency of the protocol at low and high load	6
5.	a) Explain the flooding. Consider the network as show in the figure, using flooding as the routing algorithm. If a packet sent by 1 to 6 has a maximum hope count of 3. List all the routes it will take. Also tell how many packets generated? Assume, no duplicate is discarded.  	6
	b) Discuss the network Layer Design Issues in detail.	6

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Total pages: 2

B.Tech (Computer Engineering) VIth Semester Examination 2017
Object Oriented Programming & Design
Paper Code: CEN - 603

Max. Marks: 60 Time: 3 Hours

Instruction to the candidates:
Attempt Any Two parts from each question.
Each part of the question carries 6 marks.
All programming questions should be answered using C++/ Java only.

1(a)	What is a friend function and does it violates data hiding? Justify your answer. Write a program to show how friend functions can access private data members of more than one class.	6
(b)	Answer in short. i) Why there is no return type in a constructor? ii) What is the use of set_new_handler() ? iii) What is the implicit declaration of "this" pointer? How can we use "this" pointer efficiently.	6
(c)	Write an interactive program using self-referential classes which should support insertion at end and deletion in the beginning on a single link list.	6
2.(a)	Write a program to overload subscript operator such that out of bounds memory accesses can be taken care off. Why does an overload [ ] operator can return a reference or a non-reference argument both?	6
(b)	Which are the operators which cannot be overloaded and which are the operators which cannot be overloaded using friend? Show how can you overload "<<" and ">>" operators using friend function.	6
(c)	What is the rule for writing a one argument constructor or an operator function for complete conversion between objects? Using a program show how a user defined data type is converted to a basic data type.	6

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3(a)	What is a virtual base class and what is its use? When it is necessary to write a parameterized constructor in a derived class. Show with an example to illustrate the answer.	6
(b)	What is the use of virtual destructors in C++? What is the use of pure virtual functions? Gives the rules of writing virtual functions.	6
(c)	Write a program to demonstrate the use of virtual functions in order to achieve run time polymorphism.	6
4.(a)	What is an interface? How a class can implement multiple interfaces and extend a class at the same time? Explain with suitable example.	6
(b)	What is an abstract class? How is it used to implement run time polymorphism in Java? Explain with suitable example.	6
(c	Write short notes on the following : (i) Visibility controls available in Java (ii) Java Virtual Machine (JVM) (iii) Byte Code	6
5 (a)	Differentiate between Error and Exception? What is a finally block? When and how is it used? Give a suitable example.	6
(b)	What is Multithreading? In how many ways a thread can be created in Java? Write a program to demonstrate the use of Thread class to create multiple threads.	6
(c)	Differentiate between the following with suitable example. (i) throw and throws keywords (ii) final variable, final method and final class	6

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